

Jisoo Chae

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Quality-focused Clinical Data Professional with strong experience in software quality assurance, clinical genomics, and data analysis. Proven success supporting QA/QC operations, building validated data pipelines, and collaborating with informatics and IT teams to ensure high data integrity and regulatory compliance. Currently pursuing an M.S. in Biological Data Science with hands-on training in automated testing tools, Python/R programming, and database systems. Eager to apply cross-functional experience at the intersection of genetics and software testing.

PROFESSIONAL EXPERIENCE

Molecular Technologist III | Quality Control Reagent Specialist | Molecular Technologist II
Caris Life Sciences

Phoenix, AZ
Sep 2021–May 2025

- Promoted twice within 2 years based on strong performance and technical expertise in molecular diagnostics.
- Conducted quality control and data analysis on genomic sequencing data, including whole-exome and transcriptome sequencing, ensuring clinical compliance and data integrity.
- Performed reagent QC runs using pyrosequencing and NGS, analyzing DNA methylation levels and targeted biomarkers (e.g., MSI, LOH, TMB).
- Validated gene fusions and mutation frequencies; compared current and historical data to ensure result accuracy.
- Collaborated with IT and informatics teams to optimize LIMS and automation tools, improving data processing and laboratory efficiency.
- Led training sessions for new hires on lab automation, data systems, and regulatory compliance.
- Maintained high-throughput workflows, ensured timely reporting, and supported assay development through cross-functional teamwork.

Clinical Laboratory Specialist III
Creative Testing Solution

Phoenix, AZ
Aug 2019–Sep 2021

- Conducted high-throughput clinical testing for blood donor screening and infectious disease detection.
- Performed clinical chemistry assays, including glucose, cholesterol, and enzyme tests.
- Processed ABORh typing, antigen screening, and antibody identification for transfusion services.
- Ensured compliance with FDA, AABB, CLIA, and CAP regulations for laboratory operations.
- Performed internal audits and quality control measures to verify instrument calibration and test accuracy.

EDUCATION

Arizona State University

Master of Science in Biological Data Science (GPA 3.9)

Coursework: Database Systems using SQL, Data Analysis with Python (Numpy, Matplotlib, Pandas), Statistical Analysis using R (tidyverse)

Tempe, AZ
May 2025

Arizona State University

Bachelor of Science in Biological Science | Minor in Analytical Chemistry

Tempe, AZ
May 2013

PROJECTS

The Relationship between BMI and Health

- Conducted a statistical analysis using data from the US National Health and Nutrition Examination Study (2009-2012) to evaluate the relationship between Body Mass Index (BMI) and self-reported health status categories.
- Performed ANOVA to assess BMI differences across health categories, revealing a significant association.
- Findings indicated a trend of increasing BMI with declining health status, suggesting BMI alone may not be a comprehensive indicator of overall health.
- The dataset (NHANES) includes approximately 10,000 observations, re-sampled from NHANES raw data containing 20,293 observations.
- Conducted ANOVA-F analysis due to the categorical nature of one variable and the numerical nature of the other.

The Influence of Public Health Interventions and Socioeconomics on the Spread and Severity of Infectious Diseases

- Led a group capstone research project assessing the influence of public health interventions and socioeconomic factors on the spread and severity of infectious diseases such as COVID-19.
- Conducted a systematic literature review and quantitative analysis using epidemiological datasets from various open-access databases such as Our World in Data, NCHS, WHO, and CSSE.
- Applied regression modeling and state-space transmission models to analyze the impact of vaccination rates, social distancing, and sanitation policies on infection rates and mortality.
- Investigated the socioeconomic disparities in disease transmission, demonstrating the relationship between healthcare access, demographic factors, and disease burden.
- Utilized Python and R for data processing, visualization, and statistical modeling, with Power BI for interactive data visualizations.

TECHNICAL SKILLS

- Molecular Testing: NGS Library Prep, Automated and manual DNA/RNA extraction, PCR, MGMT
- Serological Testing: ABORh testing, ELISA, Antigen screening, Antibody identification, Hemagglutination
- Regulatory Compliance: HIPAA, CLIA, CAP, FDA, and laboratory accreditation processes.
- Quality Control: Ensured data integrity through rigorous QA/QC procedures
- Programming & Automation: Python, MySQL, R, Linux Bash.
- Database & Data Analysis: MySQL, SAS, ETL processes, data reconciliation, and transformation
- Quality Assurance: SOP development, data validation, and adherence to regulatory standards.
- Healthcare Data Analysis using Microsoft Excel, Work, and Access.
- Communication: Excellent communication and interpersonal skills.

KEY QUALIFICATIONS

- **Strong Biostatistical and Analytical Foundation:**
 - Master's-level training in statistics and biostatistics with hands-on experience performing chi-square tests, t-tests, and generalized linear models. Applied statistical methods to healthcare datasets for outcome evaluation and performance benchmarking.
- **Healthcare Data Analysis Experience:**
 - Experience analyzing clinical and laboratory data with a growing focus on health claims and outcome data. Skilled in identifying cost trends, clinical effectiveness, and quality of care metrics.
- **Programming and Data Management Proficiency:**
 - Proficient in R and Python for data manipulation, statistical modeling, and visualization. Working knowledge of SQL for querying structured databases and SAS for data analysis workflows. Experienced in data cleaning, transformation, and documentation for QA and reproducibility.
- **Reporting and Communication Skills:**

- Able to write analysis plans, interpret statistical findings, and draft methods and results for reports and manuscripts. Adept at communicating technical results to cross-functional teams, including R&D, Medical Affairs, and non-technical stakeholders.
- **Process Improvement & Quality Control:**
 - Demonstrated ability to support process consistency and improve report design based on user feedback. Contributed to laboratory QA initiatives and data validation protocols to ensure data accuracy and regulatory compliance.
- **Collaborative and Detail-Oriented Mindset:**
 - Proven team player with strong attention to detail and willingness to both review others' work and accept feedback constructively. Committed to continuous learning, professional integrity, and supporting team-based problem solving.